Mind the Gap: Morphosyntactic prediction and attraction effects in Catalan-Spanish Aging Bilinguals

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Introduction. Predictive processing is considered a central mechanism supporting language comprehension [1]. Yet, in aging, its efficiency is shaped by cumulative language experience and age-related cognitive decline, which can affect the ability to generate and maintain linguistic predictions [2]. One consequence of these changes is that older adults (OAs) often struggle in contexts where successful comprehension depends on forming accurate syntactic predictions: they show more difficulty processing ungrammatical sentences (Figure 2A) and more agreement attraction errors (e.g. **The key to the cabinets were…)* than younger adults (YAs) (Figure 2B) [3, 4]. This reflects reduced predictive maintenance of morphosyntactic features. However, how cumulative language experience across the lifespan interacts with bilingualism to shape morphosyntactic prediction in aging remains underexplored [5]. In bilingual contexts, lifelong use of two languages may support prediction and reduce susceptibility to interference. Because morphosyntactic prediction relies on interference control, enhanced control in bilinguals may reduce attraction errors. In this study, we test balanced Catalan-Spanish bilinguals in Spanish to examine how age-related cognitive changes and bilingual language experience jointly modulates morphosyntactic prediction and susceptibility to agreement interference. Compared to older Spanish monolinguals [4], we predict that lifelong dual-language experience will strengthen interference control, supporting morphosyntactic maintenance and mitigating age-related increases in agreement attraction (Predictions 1-2)[5]. **Methods.** 60 YAs (Age: 18-35) and 60 healthy OAs (Age: 60-80), all early balanced L1 Catalan-L2 Spanish bilinguals, will complete a self-paced sentence reading task in Spanish. The target sample (*n* = 120) is based on an *a priori* G*Power calculation (f = 0.25; power = 0.80). 60 experimental sentences crossed Grammaticality (grammatical/ungrammatical) and attractor (match/ (sg.)/mismatch (pl.) (see Figure 1). All items use third-person verb forms and controlled subject-verb distances. Participants will also complete a brief cognitive and linguistic battery that assesses working memory (RSPAN, Keep Track), cognitive control (Stroop), and language experience (BEST, reading habits questionnaire), allowing us to explore how these individual differences relate to morphosyntactic prediction and agreement attraction effects in bilinguals. **Expected results.** (G)LMMs will be used to assess how age & bilingualism modulate susceptibility to agreement attraction, as reflected in accuracy and RTs. In terms of accuracy, bilinguals are expected to improve with increasing chronological age, suggesting experience-based gains that offset age-related cognitive decline (consistent with Prediction 1; see Figure 3A). Regarding RTs, if bilingual experience enhances interference control, age-related slowing should be attenuated specifically in attraction conditions, where predictive maintenance and inhibition are most strongly taxed, and especially when compared to previous findings in monolingual older adults [5] (See Figure 3B). With respect to agreement attraction, and consistent with Prediction 2, bilinguals are not expected to show the age-related increase in attraction effects (e.g. erroneously accepting **El pastel con los dibujos nunca *ganaron […]* "The cake$_{\text{MASC.SING}}$with the designs$_{\text{MASC.PL.}}$never won$_{\text{Vº PL.}}$[…]" as correct) that has been reported for monolinguals in previous work [3, 4]. Instead, we expect bilingual's attraction effects to either diminish with age or remain stable primarily as a function of individual differences in interference control, rather than age *per se*. Such patterns would support the view that aging entails not only decline but also potential experience-driven improvements.

References [1] Kuperberg, G. R. & Jaeger, T. F., *Lang. Cogn. Neurosci.*, 2016. [2] Harada et al., *Clin Geriatr Med,* 2013. [3] Reifegerste, J. et al., *Front. Psychol.*, 2017. [4] Cano-Sánchez, V., *UPV/EHU Unpublished dissertation*, 2025. [5] Reifegerste, J., Jarvis, R. & Felser, C., *J. Mem. Lang.*, 2020.

Figure 1. *Sample experimental item manipulating attractor number (sg. vs. pl.) and grammaticality (grammatical vs. ungrammatical sentences).*

(i) Grammatical-Mismatch (Plural subject-Singular attractor-Plural verb; PSP):

Los pasteles$_{PL}$ *con el dibujo*$_{SG}$ *nunca* ***ganaron***$_{PL}$ *un premio de repostería debido a la calidad de la masa.*

(ii) Grammatical-Match (SPP):

Los pasteles$_{PL}$ *con los dibujos*$_{PL}$ *nunca* ***ganaron***$_{PL}$ *un premio de repostería debido a la calidad de la masa.*

(iii) Ungrammatical-Mismatch (SSP):

El pastel*$_{SG}$ *con el dibujo*$_{SG}$ *nunca* ***ganaron***$_{PL}$ *un premio de repostería debido a la calidad de la masa.*

(iv) Ungrammatical-Match (SPP):

El pastel*$_{SG}$ *con los dibujos*$_{PL}$ *nunca* ***ganaron***$_{PL}$ *un premio de repostería debido a la calidad de la masa.*

The cake(s)$_{MASC.SING./PL}$ with the design(s)$_{MASC.\ SING./PL}$. never won$_{v^{e}PL}$. a price bakery due to quality dough.

Gloss: The cake with the designs never won a bakery price due to its dough quality.

Figure 2. Previous findings from Cano-Sánchez (2025) using the same stimuli in an eye-tracking study with monolinguals showing: A) Larger grammaticality effects for OAs than YAs. B) Larger agreement attraction effects for OAs in the ungrammatical-mismatch (SPP) condition.

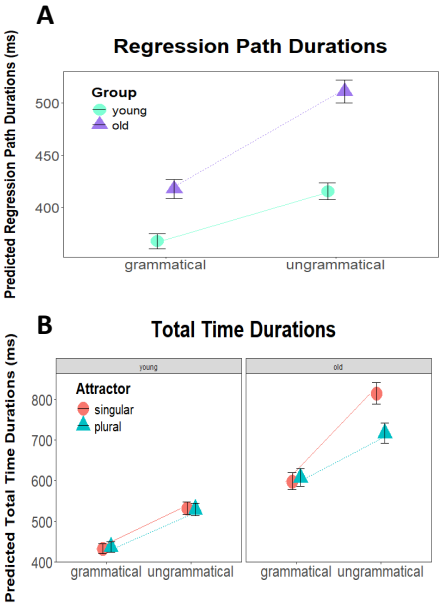
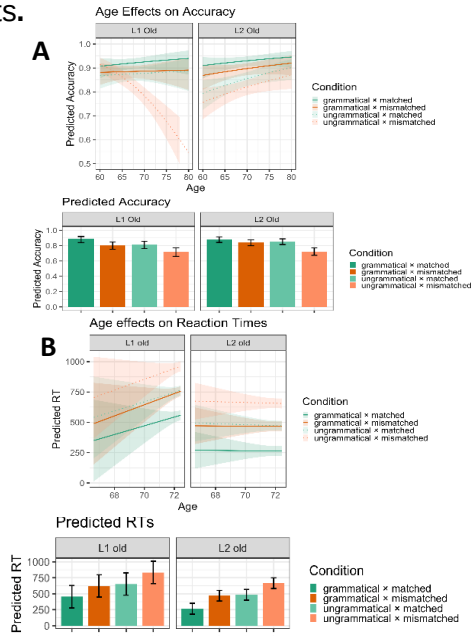
Figure 3. Predicted values for older participants based on Reifegerste et al., (2020) production study. Data were simulated using mixed-effects data-generating models (linear for RTs; logistic for accuracy), with fixed parameters specified a priori and random variability introduced at the subject and item levels: A) Accuracy rates across L1 and L2 older participants. B) Predicted RTs across L1 and L2 older participants.